

Ian Lewis Software Engineer

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Summary

- Full-stack software engineer with multiple years of experience in the JavaScript, TypeScript and Python ecosystems. Recent work includes a self-hosted observability pipeline with built-in cardinality control, a webhook inspection service deployed in a custom AWS environment, and a RAG application. Prior to software engineering, I developed a strong background in educational administration.

Technical Experience

Co-Creator, Software Engineer, *Trickl* 03/2026 – 06/2026

- Trickl** 🌐 is an open-source, scalable metrics observability platform, with built-in cardinality control measures for small to medium-sized development teams.
- Implemented a telemetry collection layer utilizing **Vector**, routing data to both a **VictoriaMetrics** backend for real-time querying capabilities and AWS **S3** for long-term storage.
 - Designed and built an **OpenTelemetry** cardinality management system that identifies label's impact on total series count and surfaces ranked reduction recommendations, applied live via hot-reload with **zero service downtime**.
 - Provisioned a production ready AWS environment using **CDK** (TypeScript), including **EC2**-backed **ECS**, **autoscaling groups**, persistent **EBS** volumes, **VPC networking**, security groups, **RDS** and **S3**.
 - Ensured production readiness by incorporating **Cloud Map** for deployed service-to-service discovery, **WAF-based API key verification** of payloads at the load balancer layer, **Cognito OIDC**-guarded **Grafana** access control, per-container **CloudWatch** logging, and managed secrets lifecycle via **AWS Secrets Manager**.
 - Designed a **React**-based visualization layer for **out-of-the-box simplicity** making **Trickl** easier to adopt.
 - Integrated an **OpenAI**-backed **agentic workflow** using a 7-class JSON classification schema and a 3-round tool-use loop to generate user-facing context.
 - Built a **Dockerized** local development environment and **CLI** to improve setup and testing consistency across local and **cloud** environments.
 - Wrote a technical case study 🌐 documenting **Trickl**'s problem domain, tradeoffs and key engineering decisions in order to **communicate complex architectural rationale clearly**.

Software Engineer, *Self-Employed* 09/2023 – 03/2026

- The Dugout 🌐 (**Python, FastAPI, React, Vite, PostgreSQL, Pinecone, OpenAI API, Vercel, Render**). Ingested and chunked source material, generated embeddings for vector storage, orchestrated semantic search using the user query to retrieve and generate a response utilizing the OpenAI API and a token friendly model.
- RequestStore (**Node.js, Express, React, TypeScript, MongoDB**). Self-hosted RequestBin-style tool for capturing and inspecting incoming webhooks and HTTP requests. Utilized a custom AWS VPC for deployment making use of public & private subnets, AZs, internet and NAT gateways and Nginx as a reverse proxy.
- LineupCFB 🌐 (**React, JavaScript, Supabase, Edge Functions, Render, Vercel**). Live ESPN score ingestion, complex scoring logic and week advancement with historical snapshots. Deployed and actively maintained.

Non-Technical Experience

Educator & Administrator, *UK & US School Systems* 08/2013 – 09/2023

- Managed budgets, mentored early-stage teachers, designed and delivered curriculum and led staff development initiatives designed to improved instructional quality and student outcomes. UK, US

Skills

AI Engineering Claude Code, Codex, OpenAI, Mastra, Multi-Agent Orchestration, RAG, Evals	Frameworks Node.js, Express, FastAPI, React, Vite, Tailwind CSS	Cloud, DevOps & Infrastructure AWS, Docker, CDK, ECS, Linux, CI/CD, Git, VSCode, Unix, Kafka Streams	Languages Python, JavaScript, TypeScript, Ruby, SQL, HTML/CSS
Databases & Storage PostgreSQL, MongoDB, Redis, S3	Deployment Vercel, Render, Supabase, Firebase, Fly.io, Heroku, Jest	Observability Honeycomb, Open Telemetry, Grafana, VictoriaMetrics, Mimir	

Education

MBA, *Northeastern State University* 09/2021 – 12/2022

Relevant courses: Statistics & Accounting, Data Analytics, Communications, Operations Management, Data Science

BS in Education, *Kingston University* 09/2010 – 07/2013
Graduated with Honors. London, England